G M AL MAMUN

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Google Scholar: Profile Link Researchgate: Profile Link

Research Interests

Autonomous Vehicles, Connected and Automated Vehicles (CAVs), Autonomous Driving Safety, Intelligent Transportation Systems (ITS), Vehicle-to-Everything (V2X) Communication, Artificial Intelligence for Smart Transportation Systems, Transportation Safety and Operations.

Education

Master of Science in Traffic & Transportation Engineering Southeast University, China Expected July, 2027

(Southeast University is currently ranked 4th in Transportation Science & Technology globally according to the Shanghai Ranking)

- Thesis: "Enhancing Autonomous Vehicle Safety Through Cooperative Perception for Occluded Vulnerable Road Users "
- Supervisor: Dr. Xiaomeng Shi
- GPA: 3.87/4.00

Bachelor of Science in Civil Engineering Presidency University, Bangladesh Oct 2016 – April 2021

- Final Year Project: "Influence of Heterogeneous Vehicle Composition on Motorcycle Accidents Occurring in Dhaka City. "
- GPA: 3.16/4.00

Research Experience

Master's Researcher Jiangsu Key Laboratory of Urban ITS, Southeast University Sep 2024 – Present

- Conduct research on autonomous vehicle safety and intelligent transportation systems.
- Develop simulation frameworks using CARLA and SUMO.
- Evaluate safety metrics including Time-to-Collision (TTC), conflict counts, and travel efficiency.
- Analyze traffic and vehicle behavior data using Python and machine learning techniques.

Research Assistant Jiangsu Province Collaborative Innovation Center of Modern Urban Traffic Technologies Sep 2024 – Present

- Assisted in literature reviews and experimental design.
- Performed data analysis and visualization.
- Contributed to technical reports and research publications.

Publications

Journal Articles

1. Das, S., Hossain, M. M., Rahman, M. A., Kong, X., Sun, X., **Al Mamun, G. M.** (2023). Understanding patterns of moped and seated motor scooter (50 cc or less) involved fatal crashes using cluster correspondence analysis. *Transportmetrica A: Transport Science*, 19(2), 2029613. Taylor & Francis. **Q1 (Scopus / Web of Science)**

2. Manik, M. H., **Al Mamun, G. M.**, Hoque, E., Kar, O. (2023). The effects of traffic congestion and a way out on circular intersections: A case study of 2 No. Gate (Soloh Sohor) area of Chattogram Metropolitan City. ***Journal of Transportation Systems***, 8(3), 7–17.

Conference Papers

1. **Al Mamun, G. M.**, N. Shiwakoti, X. Shi (2026). A hybrid control framework for safe autonomous vehicle interaction with vulnerable road users in urban driving simulations. *Transportation Research Board (TRB) Annual Meeting*, Washington D.C., USA — **one of the top-ranked and most prestigious international conference in transportation engineering**.
2. **Al Mamun, G. M.**, Nazmul Haque (2023). Influence of heterogeneous vehicle composition on motorcycle accidents occurring in Dhaka City. *7th International Conference on Engineering Research, Innovation, and Education.*
3. **Al Mamun, G. M.**, Ahmed Sajid Hasan, Kaniz Roksana, Asmay Pervin (2022). Impact of four-lane bridges on mobility of traffic flow in Dhaka–Chittagong Highway. *6th International Conference on Civil Engineering for Sustainable Development.*
4. Most Farjana, Khadiza Akter Ratna, Mohammad Abu Jafor Marup, Sharmin Mahamud, **Al Mamun, G. M.**, Shaibal Barua, Md Tahatul Azid, Fatima Akter Dina (2024). Passenger satisfaction level on bus services in Chittagong City using multiple linear regression. *7th International Conference on Civil Engineering for Sustainable Development. Journal of Transportation Systems.*

Ongoing Research Projects

Enhancing Autonomous Vehicle Safety Through Cooperative Perception for Occluded Vulnerable Road Users

- Developed a CARLA-based simulation framework for evaluating autonomous vehicle safety in occluded scenarios.
- Designed a cooperative perception architecture using V2X communication and multi-sensor data fusion.
- Assessed detection performance using mAP, precision, recall, and Time-to-Collision (TTC) metrics.
- Evaluated safety improvements for pedestrians and bicyclists under varying traffic conditions.

A Rule-Based and Reinforcement Learning Hybrid Framework for Safe Autonomous Vehicle Interaction with Vulnerable Road Users in Urban Driving Simulations

- Developed a hybrid framework combining rule-based control and reinforcement learning for autonomous driving.
- Simulated interactions among autonomous vehicles, pedestrians, bicyclists, and human-driven vehicles in SUMO.
- Evaluated safety and operational performance using TTC, conflict frequency, and collision metrics.
- Compared hybrid and conventional control strategies in urban traffic environments.

An Explainable Machine Learning Framework for Analyzing Motorcycle Crashes in Heterogeneous Urban Traffic

- Developed machine learning models to identify factors associated with motorcycle crash risk.
- Applied explainable AI techniques to interpret model predictions and feature importance.
- Analyzed roadway, traffic, and behavioral factors influencing crash severity.
- Generated data-driven recommendations for motorcycle safety improvement.

A Hybrid Cooperative–Noncooperative Game-Theoretic Framework for Modeling Motorcycle Behavior in Mixed Traffic

- Developed a game-theoretic framework for modeling motorcycle behavior in heterogeneous traffic.
- Incorporated cooperative and noncooperative decision-making for lane changing and gap acceptance.
- Simulated motorcycle–vehicle interactions under varying traffic conditions.
- Evaluated impacts on traffic efficiency, safety, and behavioral stability.

Mentorship

- Mentored undergraduate students on transportation simulation projects.
- Guided project development, data analysis, and report writing.

Guest Lecture

- Introduction to Traffic Simulation Using SUMO: A Case Study of Heterogeneous Traffic in Bangladesh, Department of Civil Engineering, North Western University, Bangladesh, 2024.

Service and Outreach

- Peer Reviewer for academic journals and international conferences.
- Volunteer in STEM outreach activities.

Professional Development

- CARLA Simulator Workshop.
- SUMO Traffic Simulation Training.
- Machine Learning Certification.
- Academic Writing Workshop.
- Research Methodology Training.

Awards and Honors

- CSC Scholarship (Full-funded Chinese government scholarship) for Master's in Transportation Engineering, from September 2024 to July 2027.

Technical Skills

Programming Languages: Python, C++

Autonomous Vehicle and Traffic Simulation: CARLA Simulator, SUMO, TESS NG, MathWorks RoadRunner

Artificial Intelligence and Machine Learning: Machine Learning, Deep Learning, Reinforcement Learning, PyTorch, TensorFlow, Scikit-Learn

Data Analysis and Visualization: OriginPro, SPSS

Transportation and Engineering Software: ArcGIS (Basic), AutoCAD

Academic and Research Tools: LaTeX, Zotero, Microsoft Visio, Microsoft Office

Language Proficiency

- **English:** Preparing for IELTS Academic (Scheduled: August 2026)
- **Proficiency Level:** Professional working proficiency (academic study, research and publications in English)
- **Chinese:** Basic proficiency
- **Native Language:** Bangla

References

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